

TABLE 2: COMPREHENSIVE BREAKDOWN OF THE GVI CALCULATION

Product Specification	Parameter Change Wt	Power Driver	CPU	Connectivity	Enclosure	Layout Plan	Data Storage	Warranty	Env. Norm	Trade Practice	Current Sensor	Volt Sensor	Temp./Humid Sensor	Weather Sensors	Cloud Storage	Analytics
Operating voltage	6	3														
Max. current	21	3														
Controller voltage	9		9													
NEMA Std	28				6											
User interface	15			3												
Connectivity	9															
Size	15				6											
Aftersales service	36					3										
Weight	15	1			3											
Cost	36		1	1			1	1								
Safety margin	25							3								
Parts env. norm	13								1							
Labour norm	1								1	3						
Sustainability norm	6								3	1						
Current sense	33		3								6					
Voltage sense	33		3									3				
Temperature sense	9		1										3			
Humidity sense	9		1										3	6		
Data storage tech	15						3								6	
Data storage size	33						1								3	1
GVI		96	333	81	303	108	114	111	32	9	198	99	54	54	189	33
Rank		9	1	10	2	7	5	6	14	1	3	8	11	12	4	13

of the GVI calculation, see Table 2. Notably, the CPU and enclosure emerge as pivotal components with a substantial likelihood of future impact and a considerable challenge in terms of redesign. Hence, these are the segments where accommodating future changes is paramount.

Coupling index. In practical scenarios, tweaking a design component often necessitates alterations in its interactions with other components—termed coupling. While GVI provides valuable insights, it does not convey coupling information. To discern coupling levels, we delve into part interrelationships, a process requiring distinct tools. Complex systems benefit from mathematical approaches and algorithms. For relatively straightforward products, a relationship graph suffices.

Begin with the highest GVI component and identify its interconnected components. Robust interfaces, entailing major changes with main component alterations, are depicted

with thick lines. Conversely, weak interfaces, demanding minimal adjustments, are represented by thin lines.

Modules. Our design predominantly hinges on modular components, minimising complex interconnections. Designs replete with intricate interlinked components benefit from consolidation into distinct modules.

In our setup, we will form a separate module for power driver relays, voltage, and current sensors. This modularisation streamlines power rating changes for the charge controller. The weather sensor, an optional item positioned away from the controller, follows a similar approach.

A distinct module encompasses cloud storage and advanced analytics for the charge controller, facilitating upgrades sans hardware modifications.

Furthermore, the CPU's limited memory capacity prompts the pru-

dent use of an SD data card for storing large HTML and graphic files. Accompanying software adjustments are warranted.

Product variants. In our analysis, it is evident that our product comes in at least six distinct configurations:

1. Basic charge controller with battery monitoring
2. Charge controller with simple weather monitoring
3. Charge controller and weather station combo
- 4, 5, and 6. All three options with cloud-based analytics

However, the idea of producing and managing all six variants raises a few challenges. For one, it could quickly turn into a logistical headache. Distributors would need to handle a larger inventory, and this might lead to situations where customers cannot find the exact product they desire.

Moreover, taking on such a plan might also impact our profits nega-